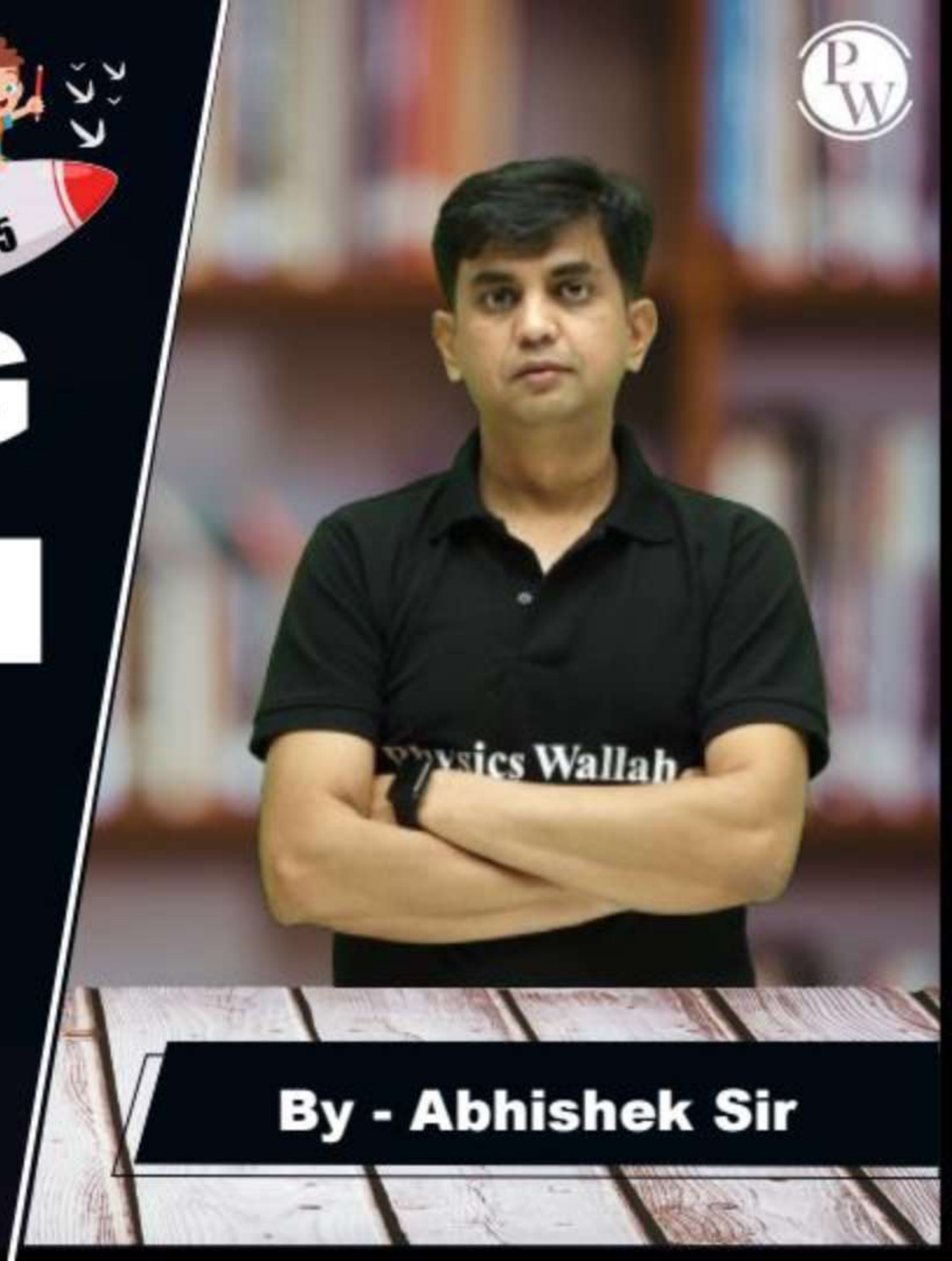
CS & IT ENGINEERING

Computer Network

Transport Layer

Lecture No. - 10



Recap of Previous Lecture

Topic TCP Congestion Control

Topics to be Covered

Topic TCP Timers

ABOUT ME

Hello, I'm Abhishek

- GATE CS AIR - 96
- M.Tech (CS) - IIT Kharagpur
- 12 years of GATE CS teaching experience

Telegram Link : https://t.me/abhisheksirCS_PW

@ABHISHEKSIRCS_PW

#Q. Which of the following statements are TRUE?

(S1) TCP handles both congestion and flow control TRUE

(S2) UDP handles congestion but not flow control FALSE

(S3) Fast retransmit deals with congestion but not flow control TRUE

(S4) Slow start mechanism deals with both congestion and flow control FALSE

[GATE-2008, 2-Mark]

(A) S1, S2 and S3 only

(B) S1 and S3 only

(C) S3 and S4 only

(D) S1, S3 and S4 only

Ans: B

#Q. Consider the following statements.

I. TCP connections are full duplex. TRUE (Point to Point)

II. TCP has no option for selective acknowledgment FALSE

III. TCP connections are message streams. FALSE

Byte stream

[GATE-2015, Set-3, 1-Mark]

(A) Only I is correct

(B) Only I and II are correct

(C) Only II and III are correct

(D) All of I, II and III are correct

Ans: A

Topic : TCP Segment Structure

0	16	31
Source Port No. (16 bit)	Dest. Port No. (16 bit)	
Sequence Number (32 bit)		
Acknowledgement Number (32 bit)		
HLEN	URG ACK PSH RST SYN FIN	Window Size (16 bit)
Checksum (16 bit)		Urgent Pointer (16 bit)
Options [Optional Header]		
—//— Payload —//—		

TCP Header (5 to 15 word) (20 to 60 byte)

TCP Base Header 5 word (20 byte)

0 to 10 word (0 to <40 byte)

HLEN →

Topic : Checksum

→ Checksum (16-bits)

→ Used to detect errors over the entire TCP segment

→ Error detection in TCP is compulsory [Unlike UDP]

(Pseudo)
Psudo Header
TCP Header
Payload

Topic : Urgent Pointer

→ Urgent Pointer (16-bits)

→ Valid only if URG (urgent) flag is set (on)

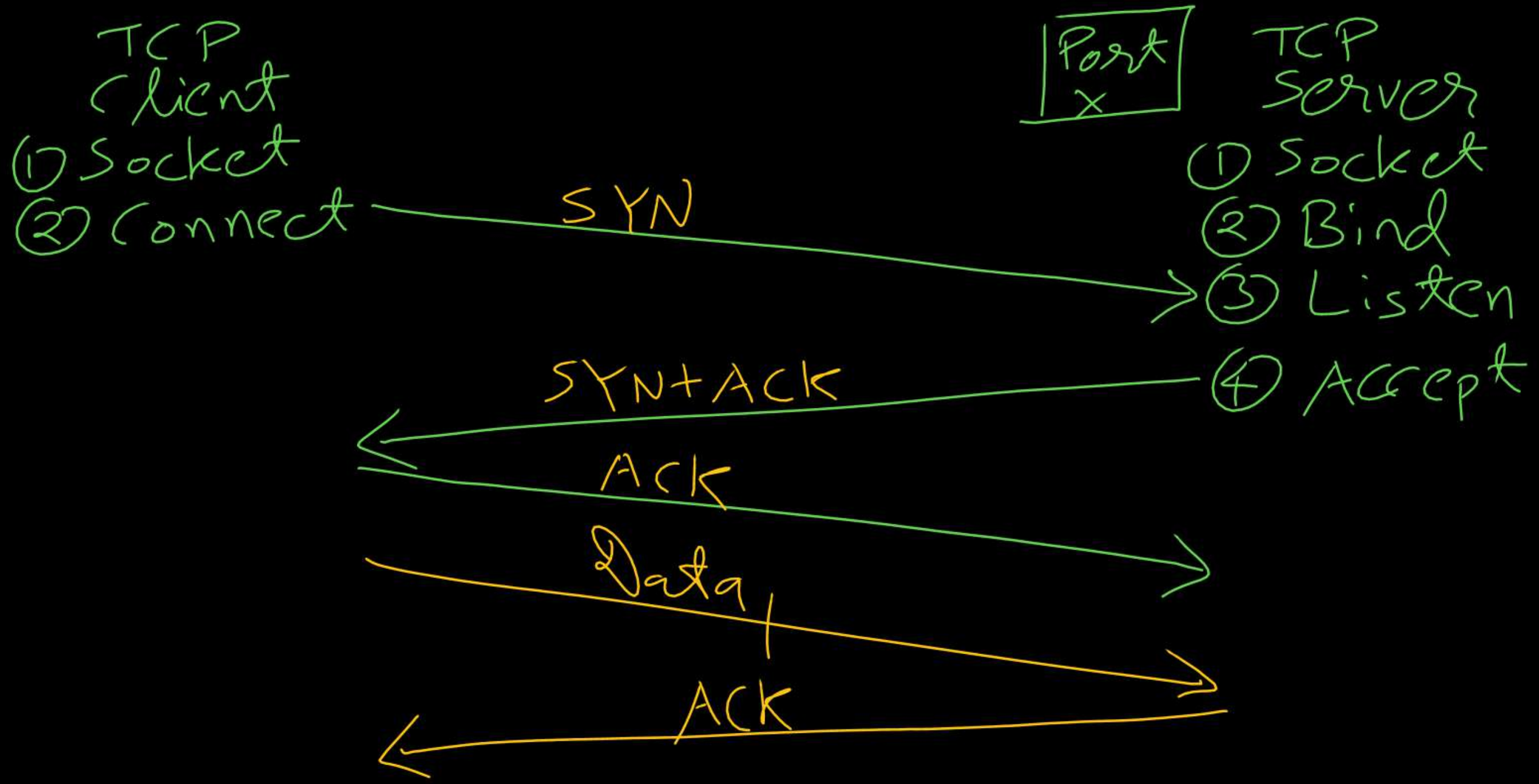
→ Defines the number to be added to the sequence number to determine the last urgent byte in the payload

Topic : Reset Flag

→ TCP reset (RST) flag

→ Used to terminate TCP connection

→ When TCP host receives a segment with the RST flag set to 1, it closes the connection (without 4-way handshaking)


TCP
Client
1 Socket
2 Connect
Port
x
TCP
Server
1 Socket
2 Bind
3 Listen
4 Accept
SYN
SYN+ACK
ACK
Data
ACK

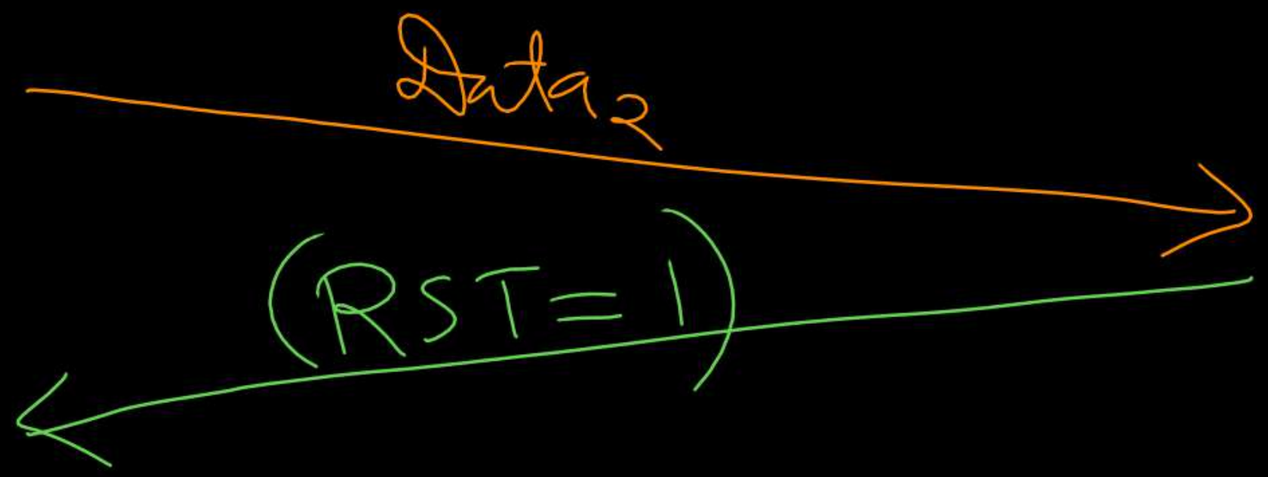
TCP Client

Server Crash

Port ✗

TCP Server

1. Socket
2. Bind
3. Listen



#Q. Which one of the following statements is FALSE?

[GATE-2004, 1-Mark]

(A) TCP guarantees a minimum communication rate FALSE

(B) TCP ensures in-order delivery TRUE

(C) TCP reacts to congestion by reducing sender window size TRUE

(D) TCP employs retransmission to compensate for packet loss TRUE

Ans: A

TCP Disadvantage :-

→ No guarantee on min^m communication rate

#Q. Packets of the same session may be routed through different paths in :

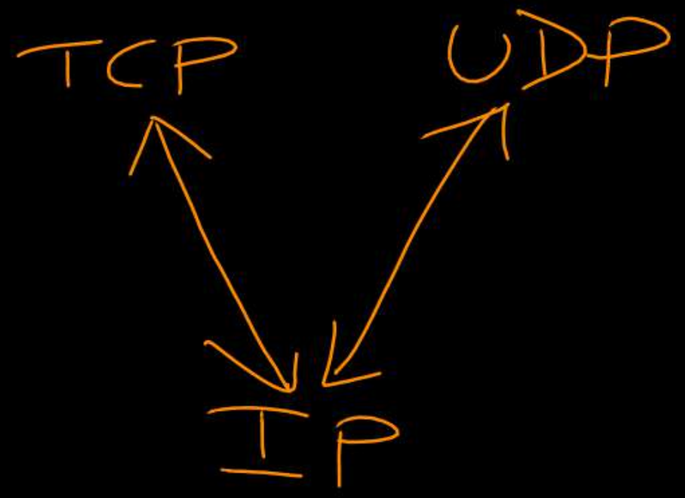
[GATE-2005, 1-Mark]

(A) TCP, but not UDP

(B) TCP and UDP

(C) UDP, but not TCP

(D) Neither TCP, nor UDP



Ans : B

Topic : Bandwidth Delay Product

→ Bandwidth * RTT

→ Receiver window size field is 16-bits long

→ if bandwidth delay product is greater than $(2^{16} - 1)$ bytes than window scaling is required

* TCP, End to End Bandwidth
* TCP, RTT

No. of bits or bytes can be transmitted in one RTT = [Bandwidth * RTT]

In TCP Header, window size (16 bit)

$\Rightarrow$ Max^m space can be advertise in the receiving buffer $= (2^{16} - 1)$ bytes

$= 65535$ byte

---

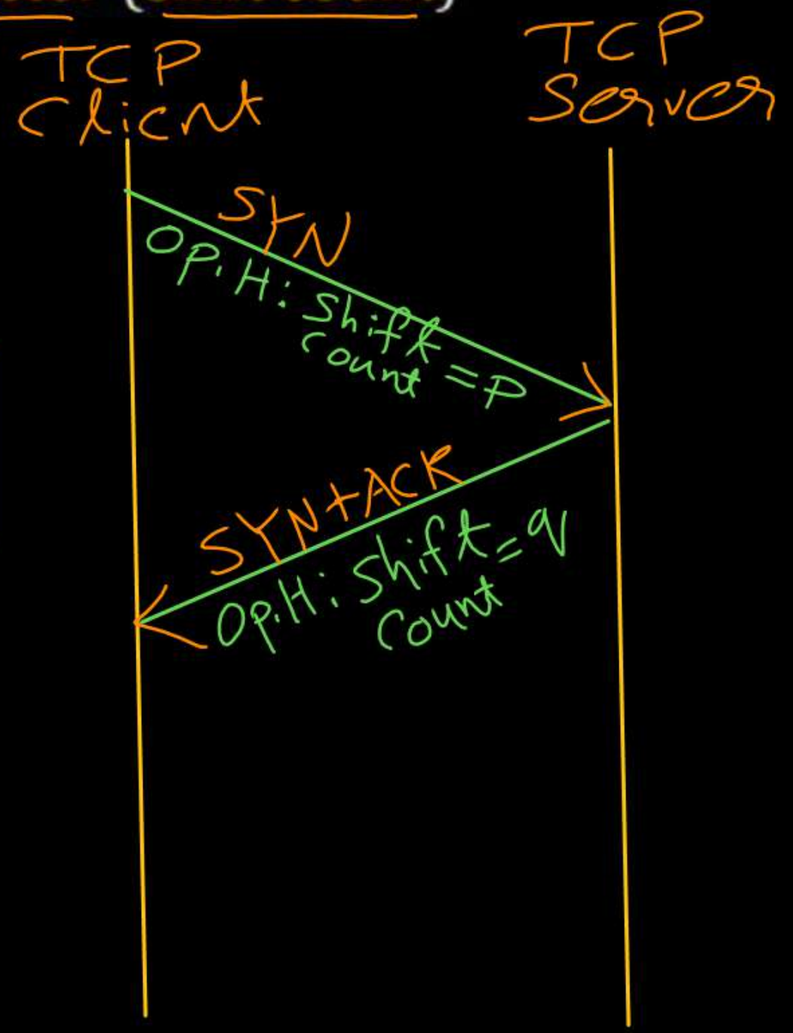
if $[(\text{Bandwidth} * \text{RTT}) > (2^{16} - 1)$ bytes$]$

Topic : Window Scaling

→ Both TCP client and server can advertise their scale factor (shift count) [during TCP handshake via SYN packet]

→ $\text{scale_factor} = 2^{\text{shift_count}}$

→ Available space in receiver's receiving buffer
= Advertised Window Size * scale_factor



Topic : Window Scaling

shift_count ≤ 14

→ Maximum value for shift count can be 14

→ Maximum possible available space in receiver's receiving buffer can be advertised with window scale is $(2^{16} - 1) * 2^{14}$ bytes

#Q. Assume that the bandwidth for a TCP connection is 1048560 bits/sec. Let α be the value of RTT in milliseconds (rounded off to the nearest integer) after which the TCP window scale option is needed. Let β be the maximum possible window size with window scale option. Then the values of α and β are.

[GATE-2015, Set-2, 2-Mark]

IIT-K, H.W.

(A) 63 milliseconds 65535×2^{14}

(B) 63 milliseconds 65535×2^{16}

(C) 500 milliseconds 65535×2^{14}

(D) 500 milliseconds 65535×2^{16}

Topic : TCP Timer

TCP uses five timers :

1. Retransmission Timer
2. Persistence Timer
3. Keepalive Timer
4. Time-waited Timer
5. Delayed Timer

Topic : Retransmission Timer

→ Used to control a lost or discarded segment

→ TCP creates a "retransmission timer" when it sends a segment

→ TCP uses dynamic retransmission time, which is different for each connection and may be changed during connection

*In, data link layer : static retransmit time

Topic : TCP Time-out Value

SampleRTT : RTTs

→ Measured time from segment transmission until ACK receipt

→ SampleRTT will vary

EstimatedRTT :

Initially : EstimatedRTT = SampleRTT

$$\boxed{\text{EstimatedRTT} = (1 - \alpha) * \text{EstimatedRTT} + \alpha * \text{SampleRTT}}$$

where $\alpha = 0.125$

Topic : TCP Time-out Value

DevRTT :

$$\text{DevRTT} = (1 - \beta) * \text{DevRTT} + \beta * |\text{SampleRTT} - \text{DevRTT}|$$

where $\beta = 0.25$

$$\text{Time-out Interval} = \text{EstimatedRTT} + 4 * \text{DevRTT}$$

#Q. Consider the following statements about the timeout value used in TCP.

i. The timeout value is set to the RTT (Round Trip Time) measured during TCP connection establishment for the entire duration of the connection. FALSE

ii. Appropriate RTT estimation algorithm is used to set the timeout value of a TCP connection. TRUE

iii. Timeout value is set to twice the propagation delay from the sender to the receiver. FALSE [TO value > $2t_p$]

Which of the following choices hold?

[GATE-2007, 1-Mark]

IIT-K

(A) (i) is false, but (ii) and (iii) are true

(B) (i) and (iii) are false, but (ii) is true

(C) (i) and (ii) are false, but (iii) is true

(D) (i), (ii) and (iii) are false

Ans: B

#Q. Suppose two hosts use a TCP connection to transfer a large file. Which of the following statements is/are False with respect to the TCP connection?

I. If the sequence number of a segment is m, then the sequence number of the subsequent segment is always m+1.

II. If the estimated round trip time at any given point of time is t sec, the value of the retransmission timeout is always set to greater than or equal to t sec.

III. The size of the advertised window never changes during the course of the TCP connection.

IV. The number of unacknowledged bytes at the sender is always less than or equal to the advertised window

[GATE-2015, Set-1, 1-Mark]

IIT-K, H.W.

(A) III only

(B) I and III only

(C) I and IV only

(D) II and IV only

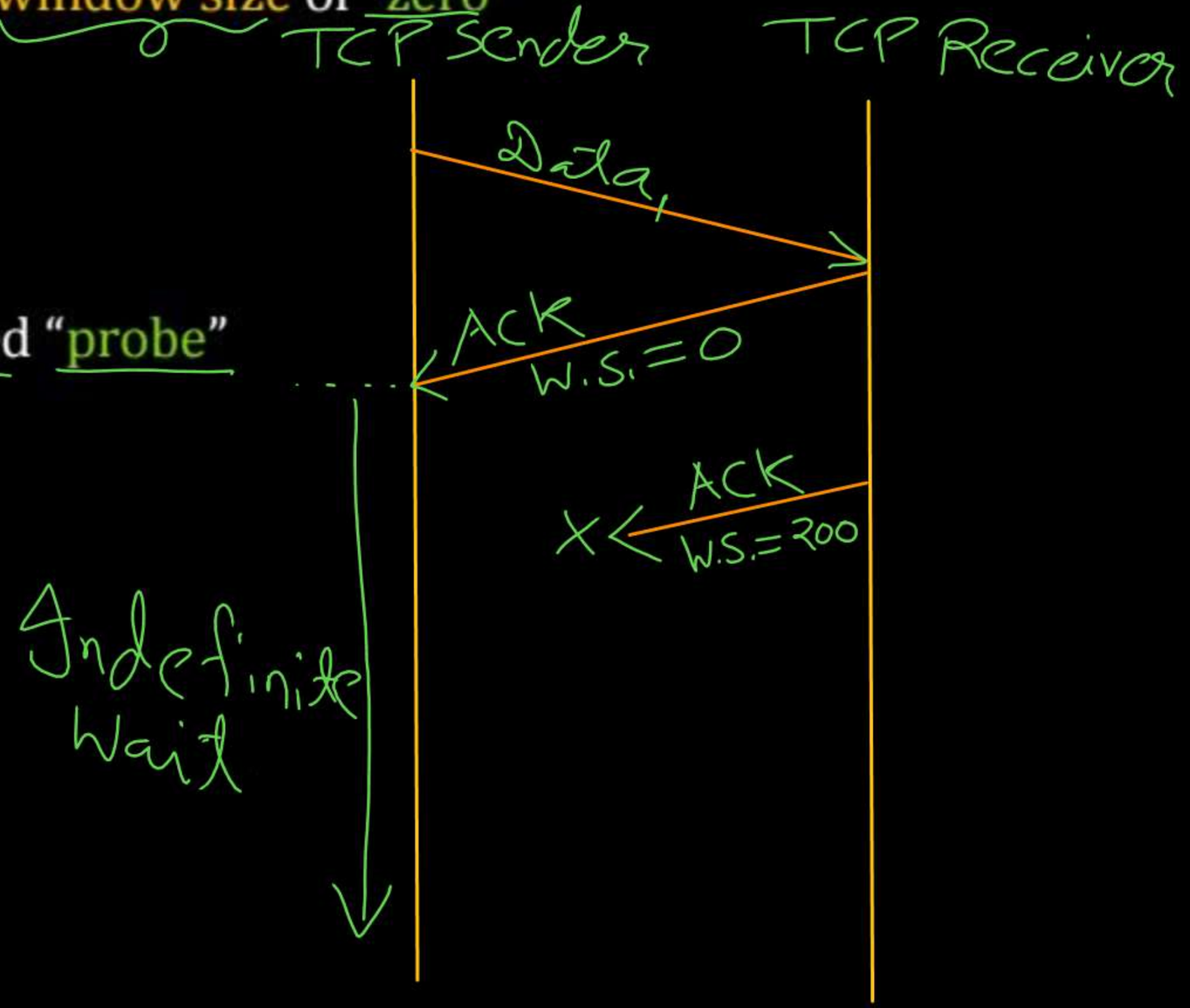
Topic : Persistence Timer

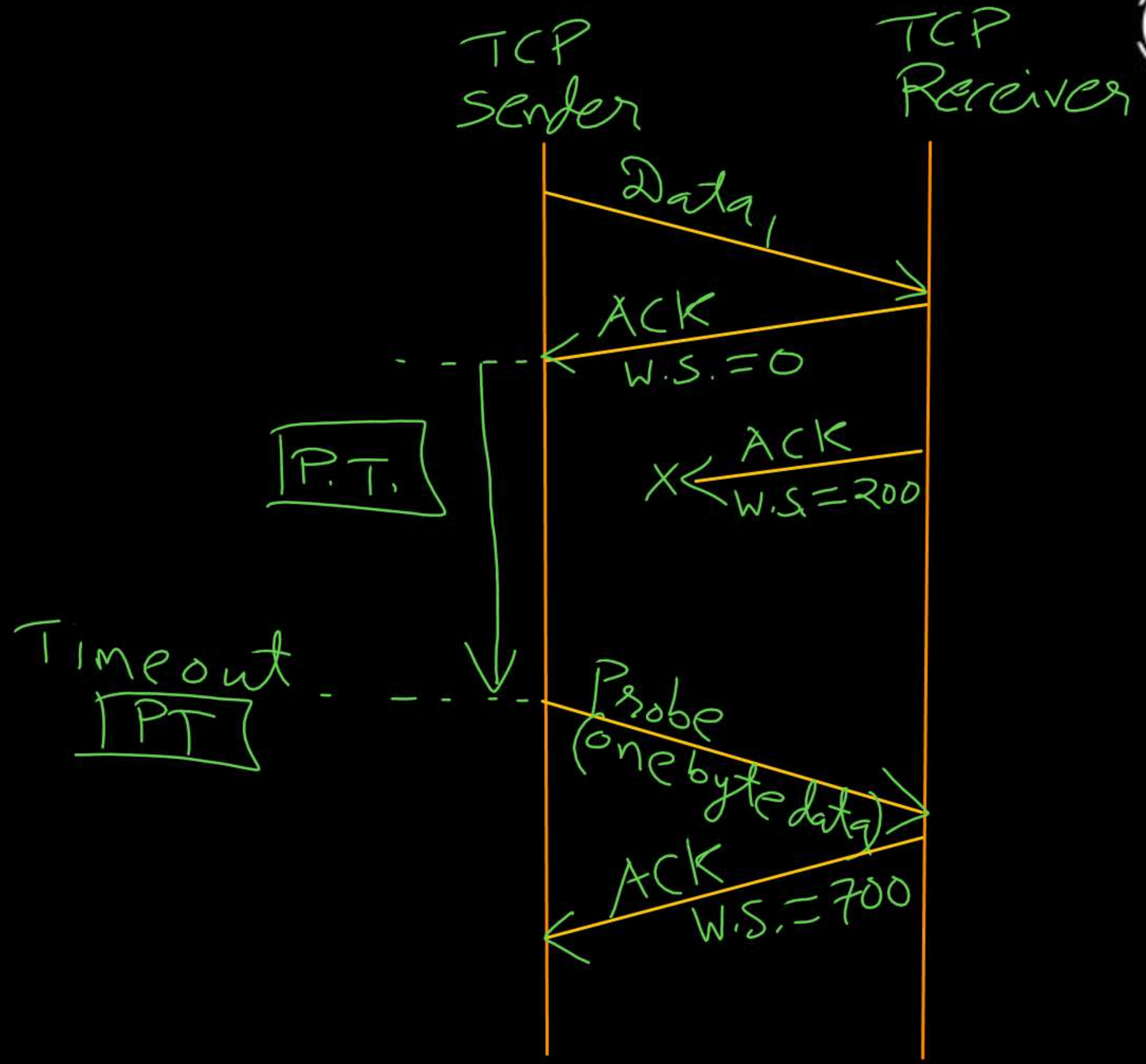
→ When TCP receives a segment with a window size of "zero"

→ TCP starts a persistence timer

→ When the persistence timer goes off, the TCP sends a special segment called "probe"

→ This prevents from deadlock




TCP Sender
TCP Receiver
Data
ACK
W.S. = 0
P.T.
ACK
W.S = 200
Timeout
PT
Probe
(one byte data)
ACK
W.S. = 700

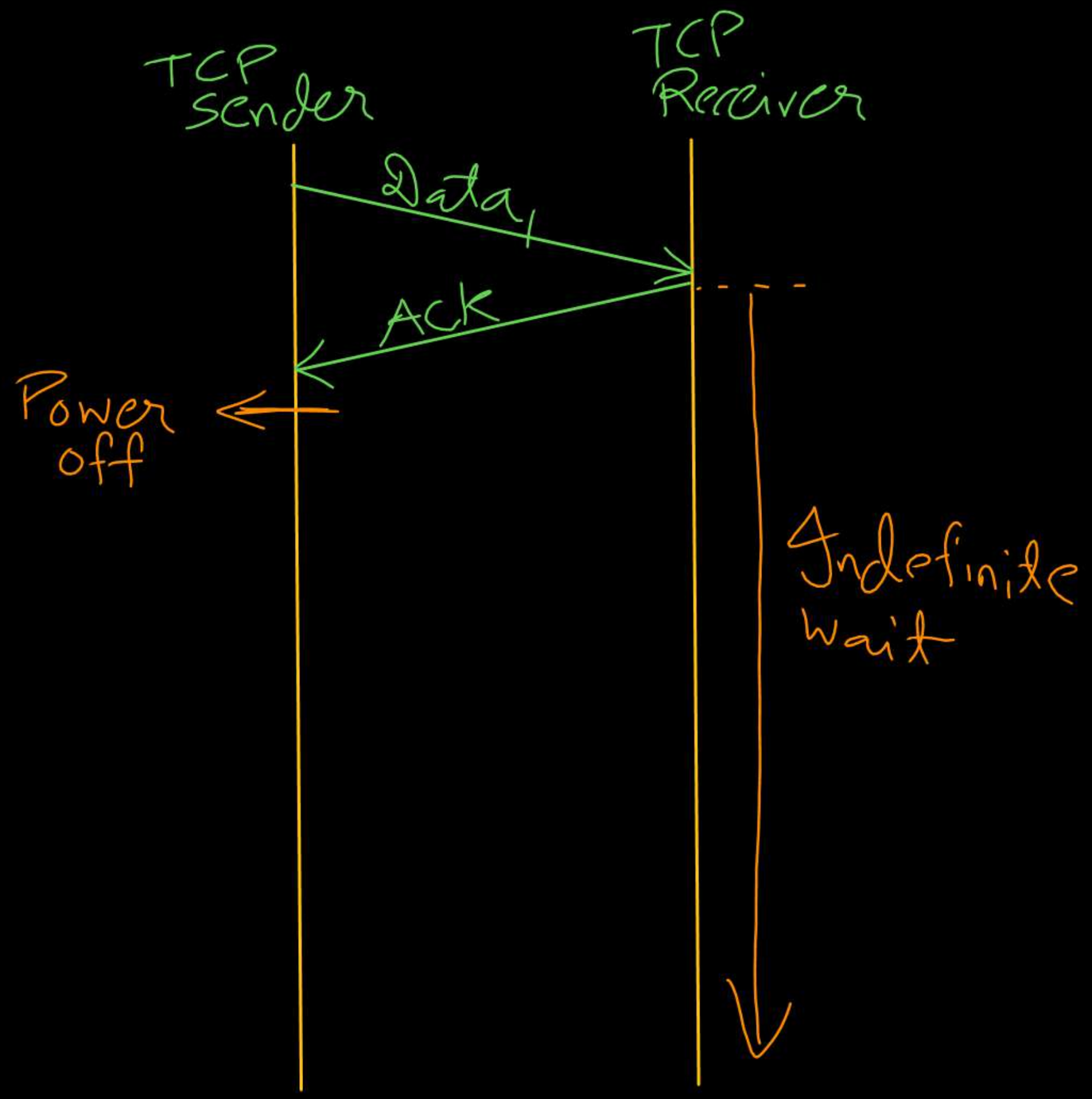
Topic : Keepalive Timer

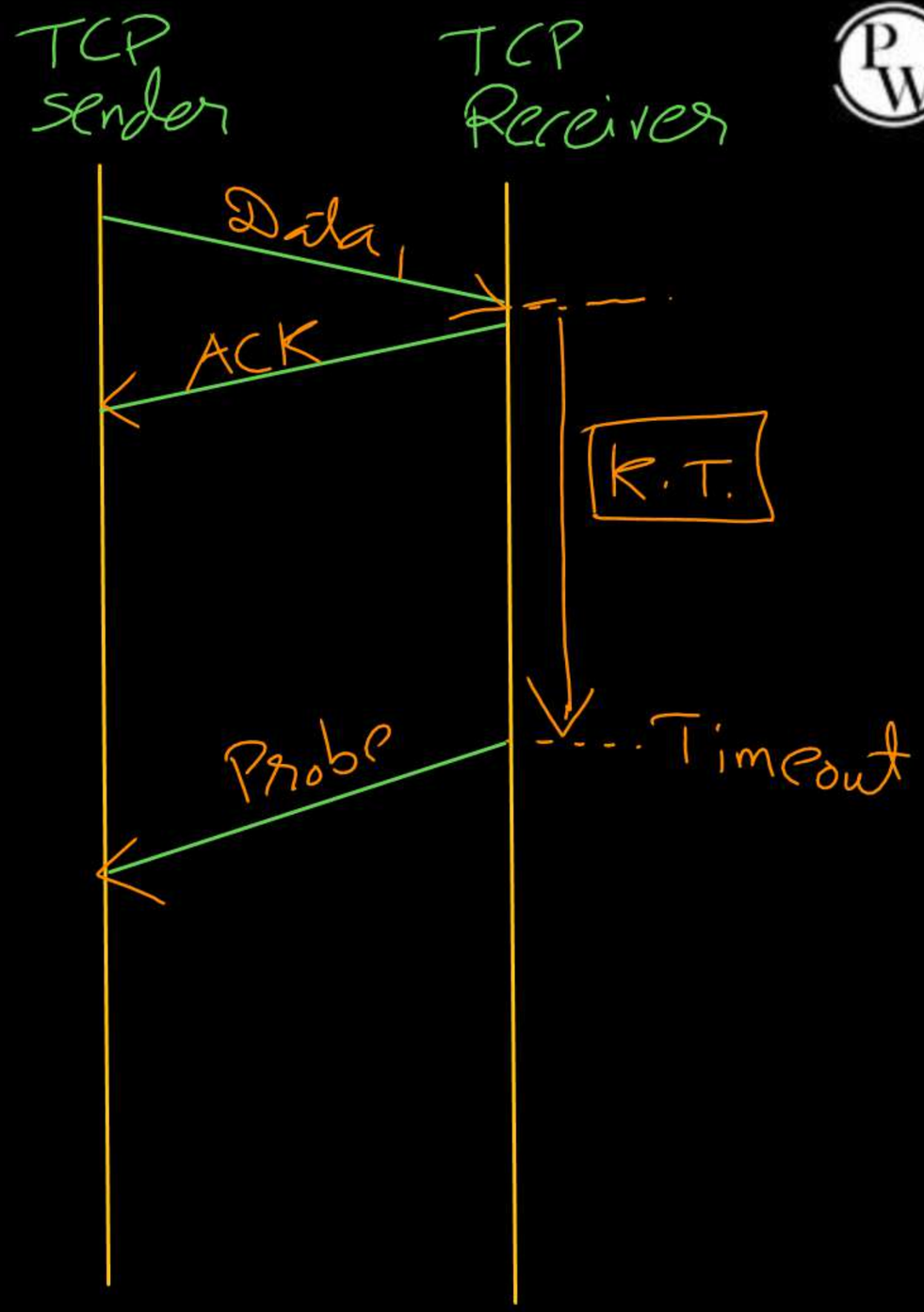
→ Prevents long idle connections between two TCPs

→ Each time TCP hears from other TCP, it reset the timer

→ if TCP does not hear from other TCP till keepalive timer time-out, the TCP sends a special segment called "probe"

→ This prevents from indefinite wait


TCP sender
TCP Receiver
Data1
ACK
Power off
Indefinite wait


TCP sender
TCP Receiver
Data1
ACK
K.T.
Timeout
Probe

[MSQ]

#Q. A TCP server application is programmed to listen on port number P on host S. A TCP client is connected to the TCP server over the network. Consider that while the TCP connection was active, the server machine S crashed and rebooted. Assume that the client does not use the TCP keepalive timer. Which of the following behaviors is/are possible?

H.W.

[GATE-2021, Set-1, 2-Mark]

(A) If the client was waiting to receive a packet, it may wait indefinitely

(B) The TCP server application on S can listen on P after reboot

(C) If the client sends a packet after the server reboot, it will receive a RST segment

(D) If the client sends a packet after the server reboot, it will receive a FIN segment

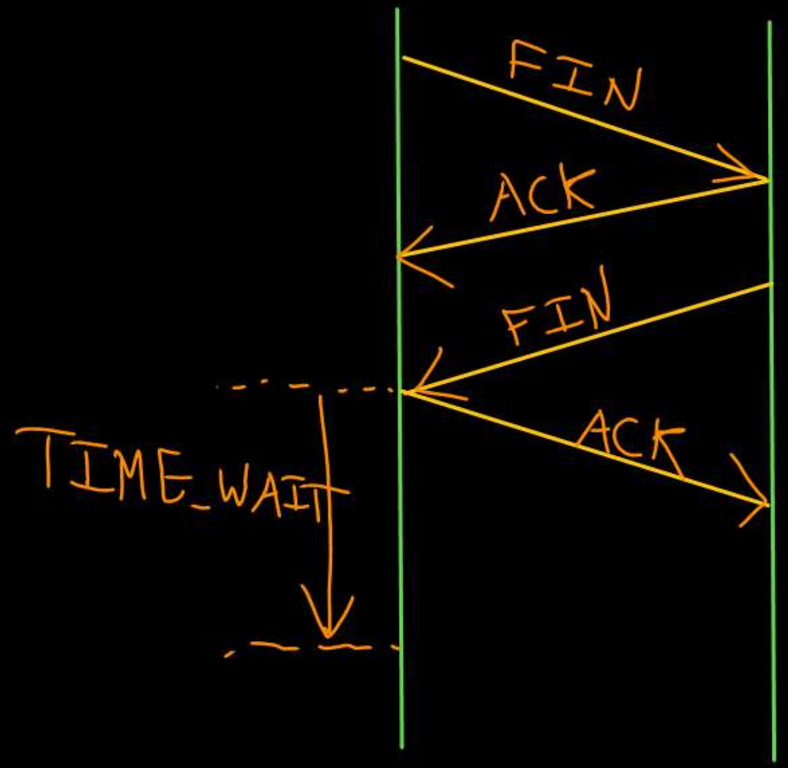
Topic : Time-waited Timer

→ Used during connection termination

→ More than twice the RTT / 2RTT OR 2MSL

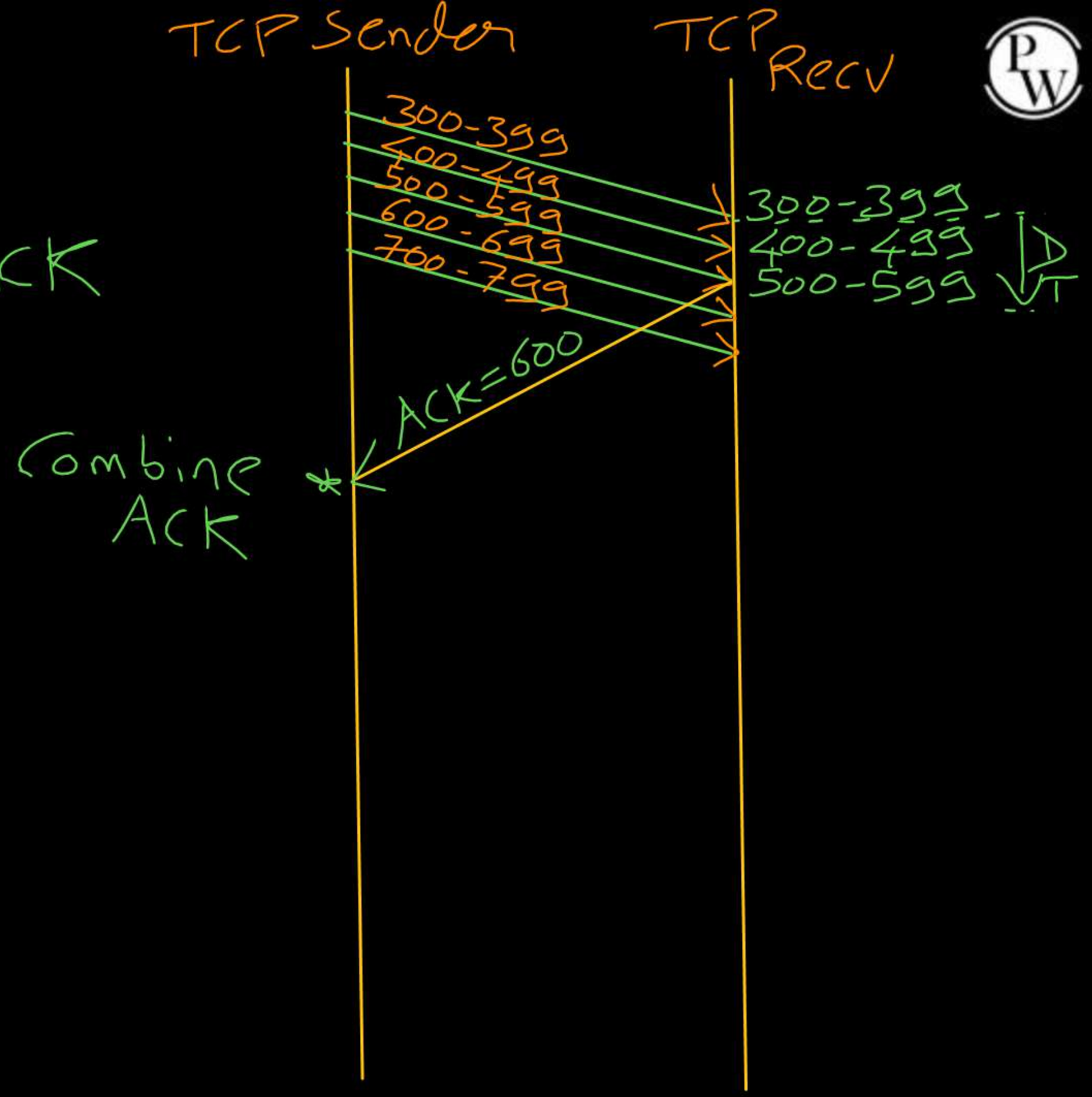
→ When TCP closes a connection, it is not really closed, until this timer time-out

→ Allow FIN segment to be received



Delayed Timer:-

* Delayed ACK used as combine ACK



Topic : TCP Optional Header

→ Also known as "Options"

→ Options are variable in size

→ Maximum size of TCP optional header is 10 words
[Words of 32 bits, i.e. 40 bytes]

→ if options size are not in words
than it uses padding [End-of-Option Option]

Topic : TCP Optional Header

→ Options field component :-

1, Option Kind (1 byte)

2. Option Length (1 byte)

3. Option Data

Topic : TCP Optional Header

Options

Topic : TCP Options

Kind	Length	Purpose
0	-	End of Option [Padding]
1	-	No Operation
2	4	(MSS) Maximum Segment Size
3	3	Window Scale (Shift count)
4	2	Selective ACK Permitted
5	10 / 18/ 24 / 36	[Selective ACKs (SACK)]
8	10	[TImestamp]

2 mins Summary

Topic TCP Timers

THANK - YOU